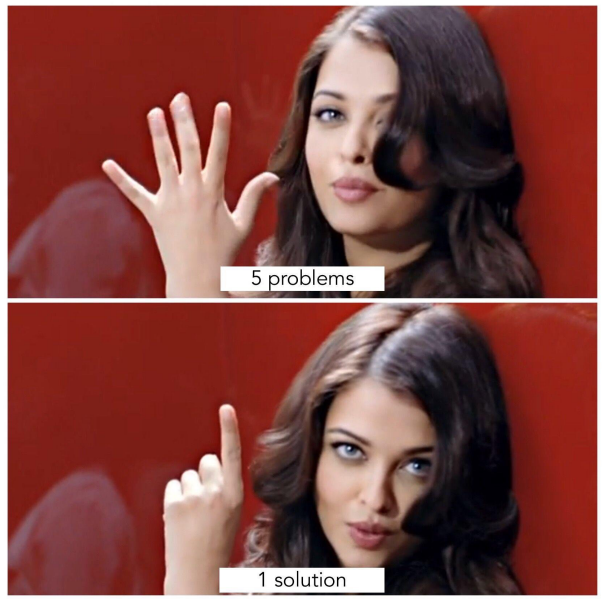
Failure of Uniqueness for Scalar Conservation Laws

One Problem, (Infinitely) Many Solutions

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Joint work with
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One equation, one initial state

Consider the Cauchy problem for a scalar conservation law:

$$u_t + \partial_x f(u) = 0, \qquad u(x,0) = u_0(x).$$

Classical solutions break down, so we must work with weak solutions

$$\iint u\varphi_t + f(u)\varphi_x \, dx \, dt = 0 \qquad \text{for all } \varphi \in \mathbf{C}^1_c(\mathbb{R} \times \mathbb{R}_+).$$

However, weak solutions are not unique, so we need a further selection criterion.

The entropy inequalities select the admissible ones.

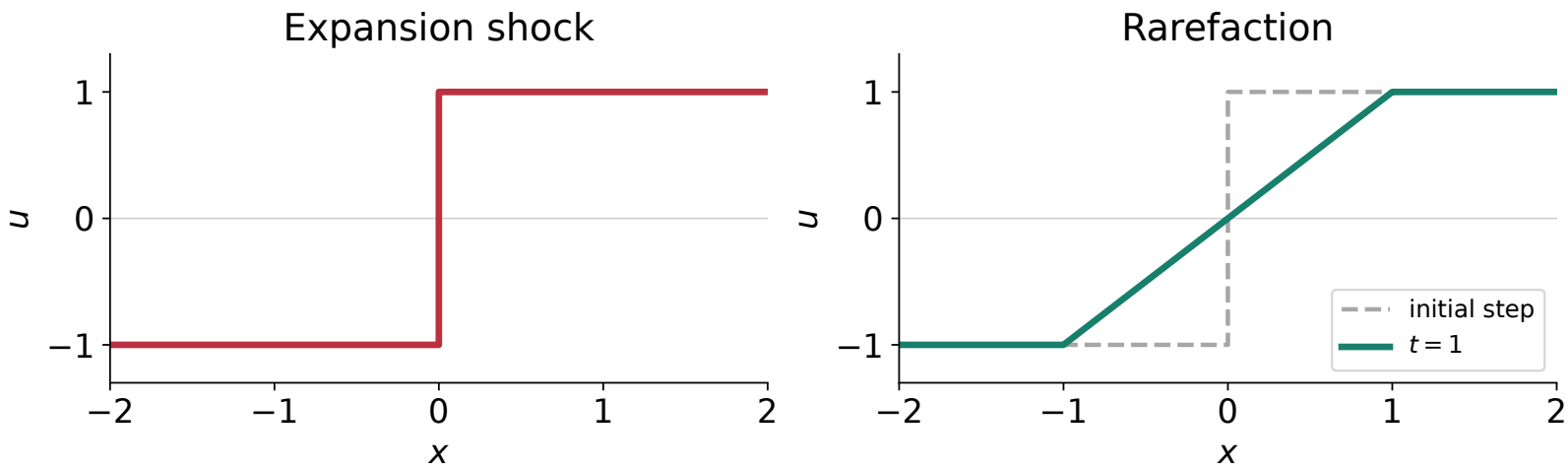
$$\eta(u)_t + \partial_x q(u) \leq 0 \text{ in distribution,}$$

where η is any convex function and $q' = \eta' f'$.

One initial state, two weak solutions

Consider the Burgers equation with step data:

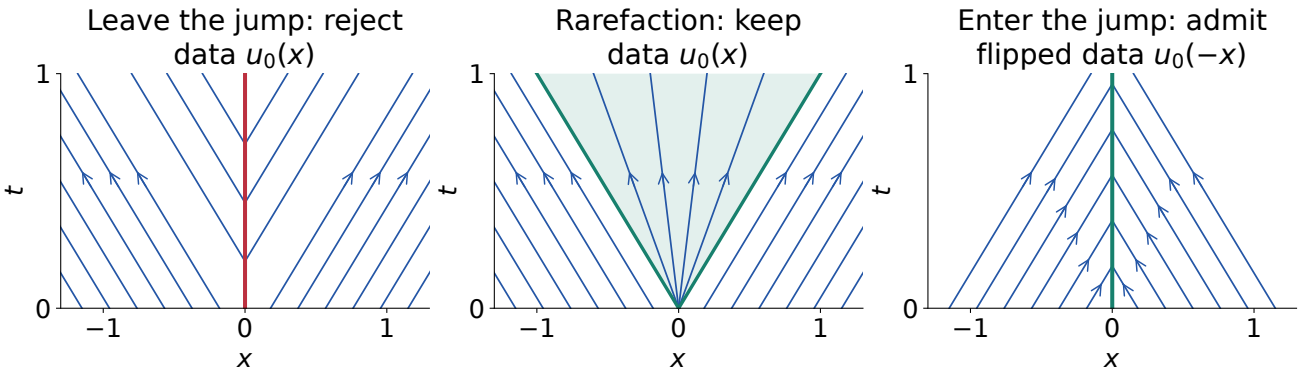
$$u_t + \partial_x(u^2/2) = 0, \qquad u_0(x) = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$



Both are weak solutions with the same initial trace.

The jump has speed $s = (u_L + u_R)/2 = 0$, while the fan is $u = x/t$.

Entropy and the Lax condition



Lax: at a shock, characteristics must enter from both sides, i.e.

$$u_L > s > u_R, \qquad s = \frac{u_L + u_R}{2}.$$

Entropy: for every constant k, we require

$$\partial_t |u - k| + \partial_x \Big[\operatorname{sgn}(u - k)(u^2 - k^2)/2 \Big] \leq 0.$$

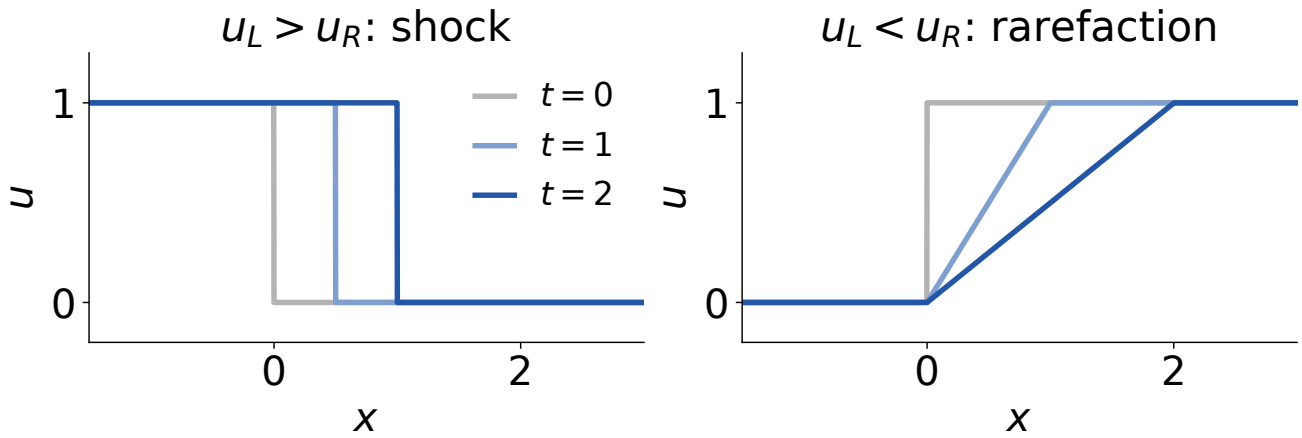
Both rule out the expansion shock and keep the rarefaction.

The Riemann problem

A Riemann problem is the Cauchy problem with a single jump:

$$u_0(x) = \begin{cases} u_L, & x < 0, \\ u_R, & x > 0. \end{cases}$$

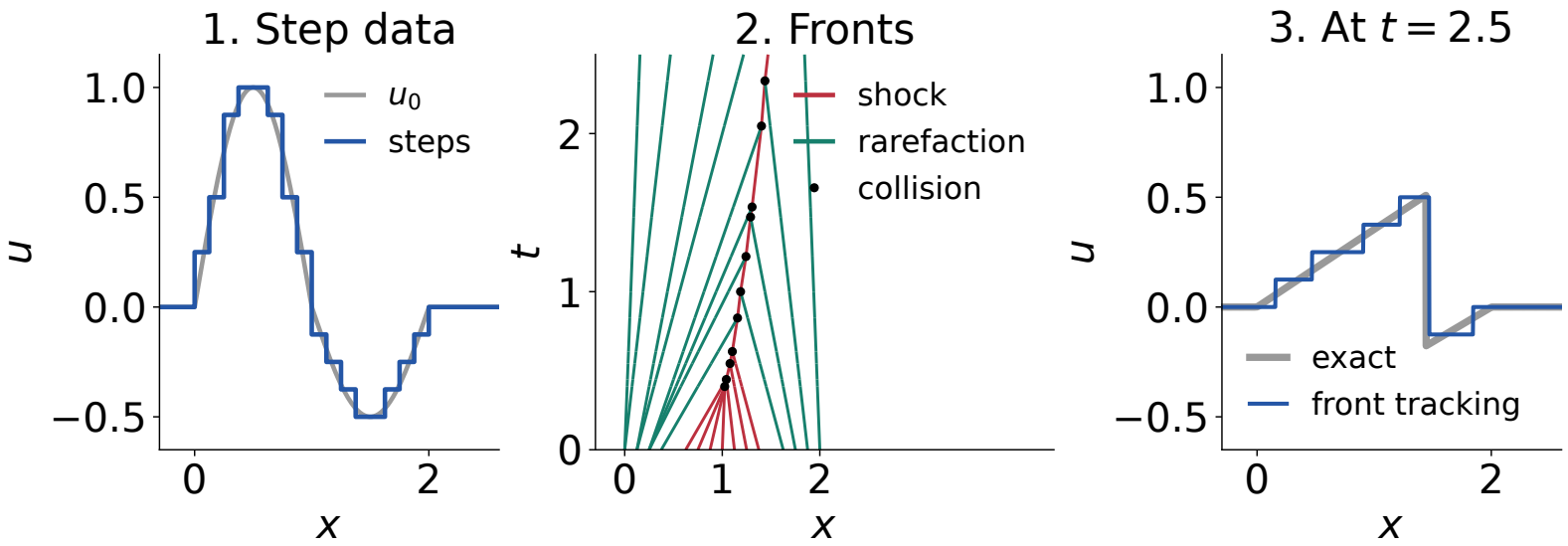
Rescaling $(x, t) \mapsto (\lambda x, \lambda t)$ leaves it unchanged, so the entropy solution has the form $u = U(x/t)$.



For Burgers: a **shock** with speed $(u_L + u_R)/2$ if $u_L > u_R$, and a **rarefaction** $u = x/t$ if $u_L < u_R$.

We have just solved two of them.

Front tracking



1. Round u_0 to a step function, with values on a grid of spacing δ.
2. Solve a Riemann problem at each jump; rarefactions become small steps.
3. Move the fronts in straight lines until two of them meet.
4. Solve the new Riemann problem there, and repeat.

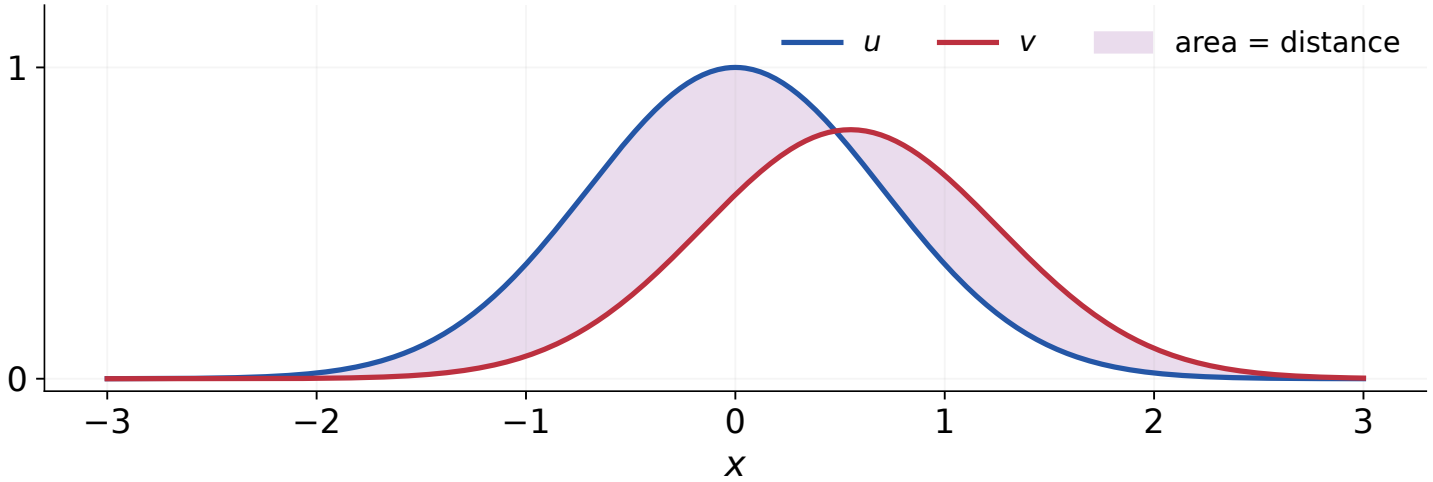
Each front is a shock or a rarefaction step; two colliding fronts leave at most one.

Burgers, $\delta = 1/8$, computed with `fronttrack`. Exact solution: Hopf–Lax formula.

Uniqueness and stability

We measure the distance between two solutions u and v by

$$D(t) = \int_{\mathbb{R}} |u(x,t) - v(x,t)|\, dx.$$



If we can show that the distance never grows, i.e. $D(t) \leq D(0)$, then the same initial data give $D(0) = 0$, and hence $u = v$.

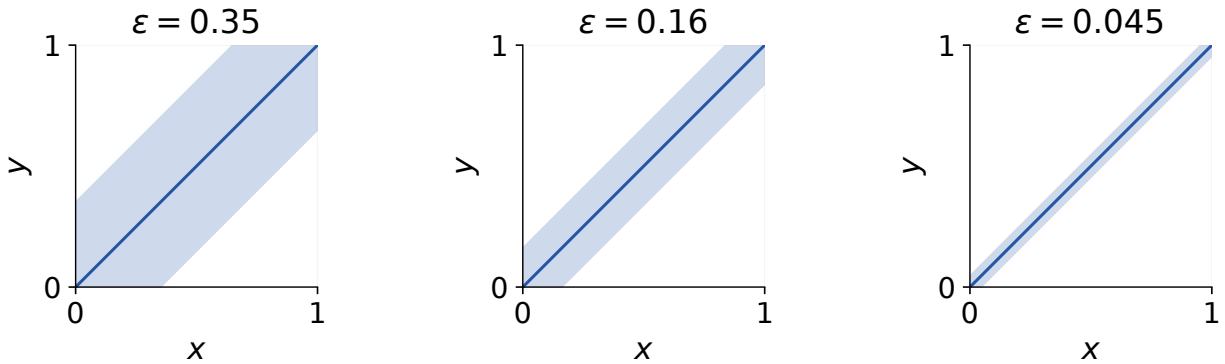
Schematic profiles. Comparison assumes integrable difference and control at infinity.

Kružkov's doubled variables

An entropy solution satisfies, for every constant k,

$$\partial_t |u - k| + \partial_x \left[\operatorname{sgn}(u - k)(f(u) - f(k)) \right] \leq 0.$$

The trick: make two copies of spacetime, and use each solution as 'k' in the other.



Adding the two inequalities and approaching the diagonal, we get

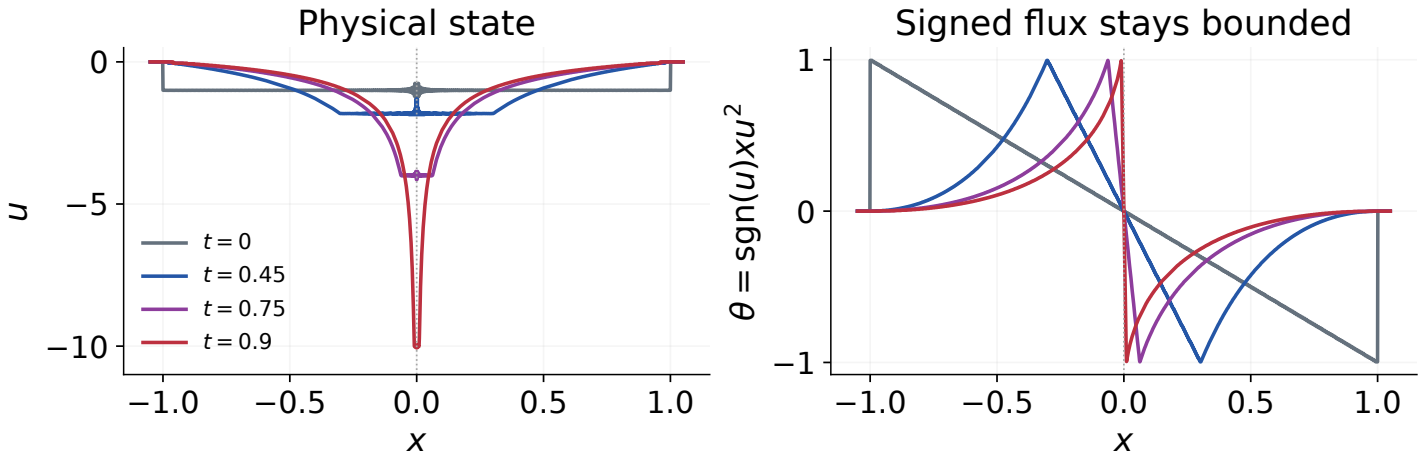
$$\partial_t |u - v| + \partial_x Q \leq 0.$$

The spatial derivative can be handled by choosing an appropriate test function.

Smooth operator

Suppose the flux depends on space, say $f(x,u) = xu^2$. The characteristics of

$$u_t + \partial_x(xu^2) = 0 \quad \text{are} \quad \dot{x} = 2xu, \quad \dot{u} = -u^2,$$



which we can solve explicitly:

$$u(t) = \frac{u_0}{1 + tu_0}, \qquad x(t) = x_0(1 + tu_0)^2.$$

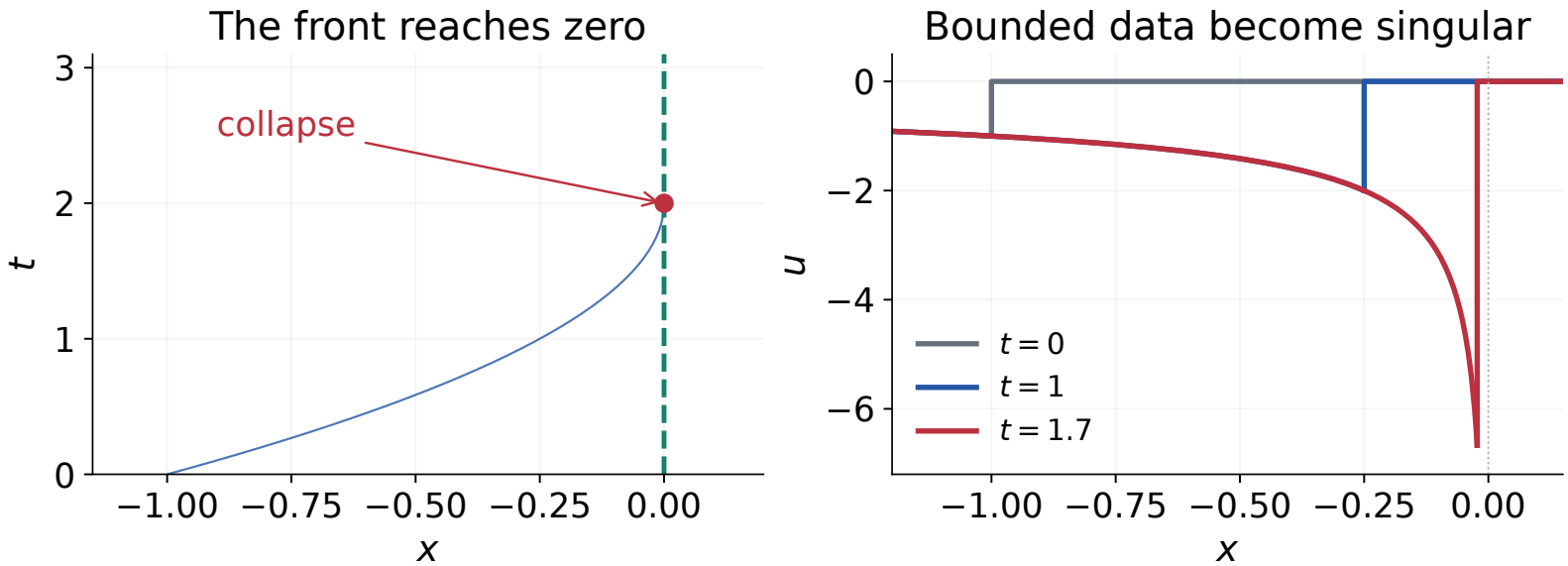
The amplitude blows up, even though the flux is conserved.

Click the figure to play : coloured profiles are time slices persisting.

The degenerate singularity

Starting from bounded data, a shock front moves along $X(t)$:

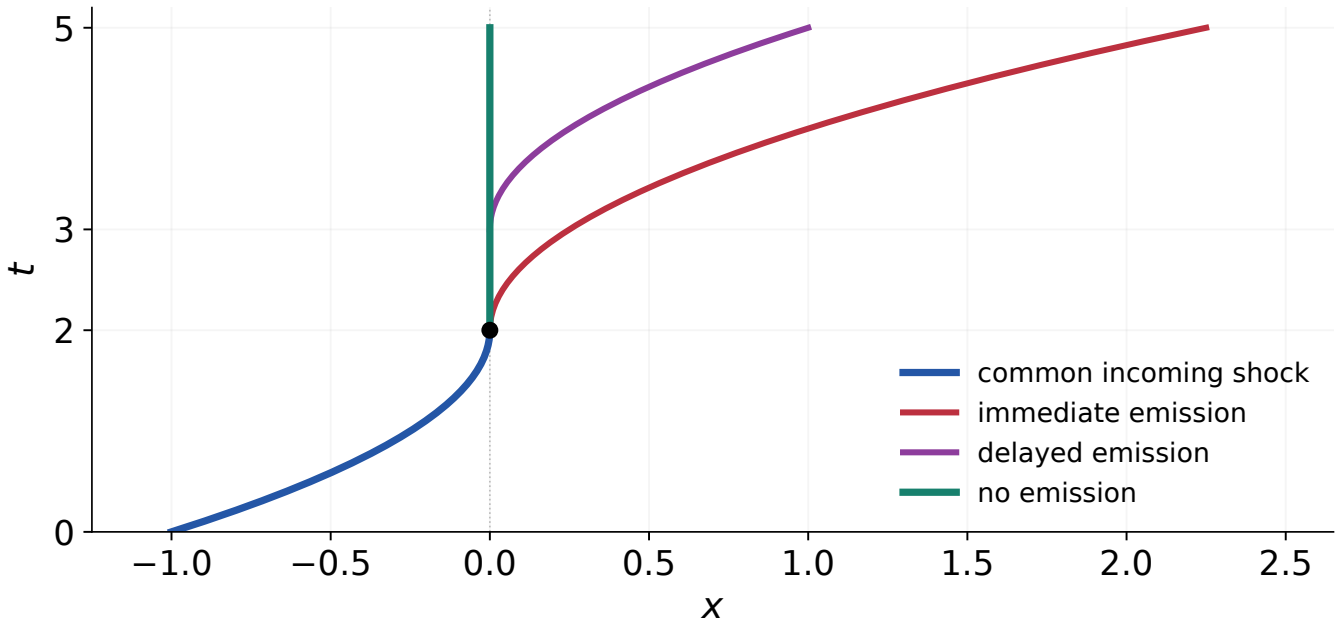
$$u_0(x) = -\frac{\mathbb{1}_{\{x<-1\}}}{\sqrt{-x}}, \qquad X(t) = -\frac{(2-t)^2}{4} \quad (t < 2).$$



At $t = 2$, the front hits $x = 0$, and the solution becomes unbounded there.

Click the figure to play the incoming front (10 seconds).

One initial state, infinitely many continuations



$$X(t) = \frac{\sigma^2 (t-\tau)_+^2}{4}, \qquad 0 < \sigma \leq 1, \quad \tau \geq 2.$$

The interface can emit a front immediately, wait as long as it likes, change the strength, or never emit at all.

Click the figure to play four continuations on one clock (15 seconds).

Hölder? I barely know her!

Where might this freedom come from? Look at the fronts themselves:

$$\dot{X} = \sqrt{|X|} \ \text{(incoming)}, \qquad \dot{X} = \sigma\sqrt{|X|}, \ \ X(\tau) = 0 \ \ \text{(emitted)}.$$

This is the classic example of non-uniqueness for ODEs,

$$\dot{y} = \sqrt{y}, \qquad y(0) = 0,$$

which is solved by $y \equiv 0$ and by $y = (t-\tau)_+^2/4$ for every $\tau \geq 0$.

Of course, each branch must still satisfy the entropy inequalities.

Shock front equations follow from the characteristic equations and formulas on the two previous slides (the incoming one also from Rankine–Hugoniot).

Entropy inequalities do not imply weak conservation

For bounded states, the entropy inequalities imply the weak equation.

For unbounded states, this implication can fail.

In fact, some unbounded data admit no weak solution at all:

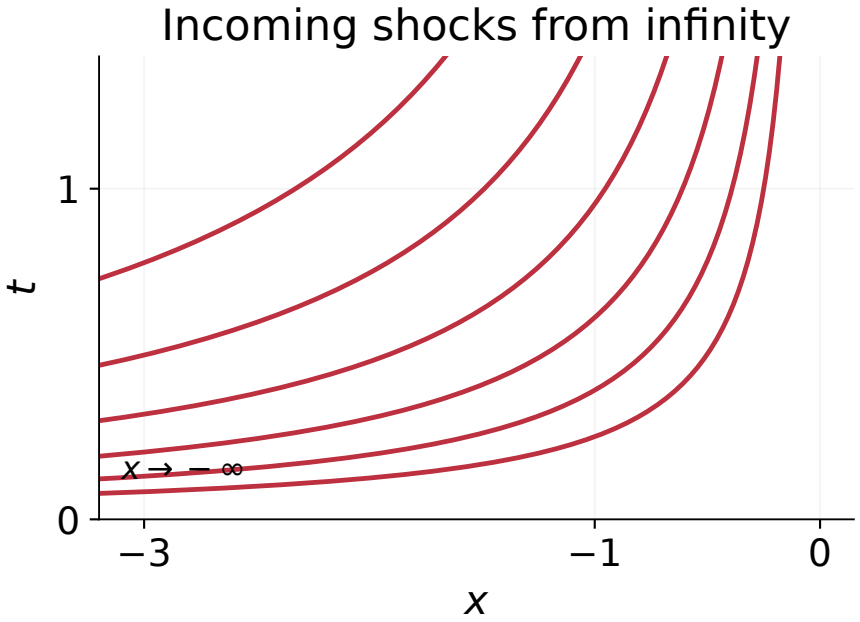
$$u_0(x) = -\frac{\mathbb{1}_{\{x>0\}}}{\sqrt{x}}.$$

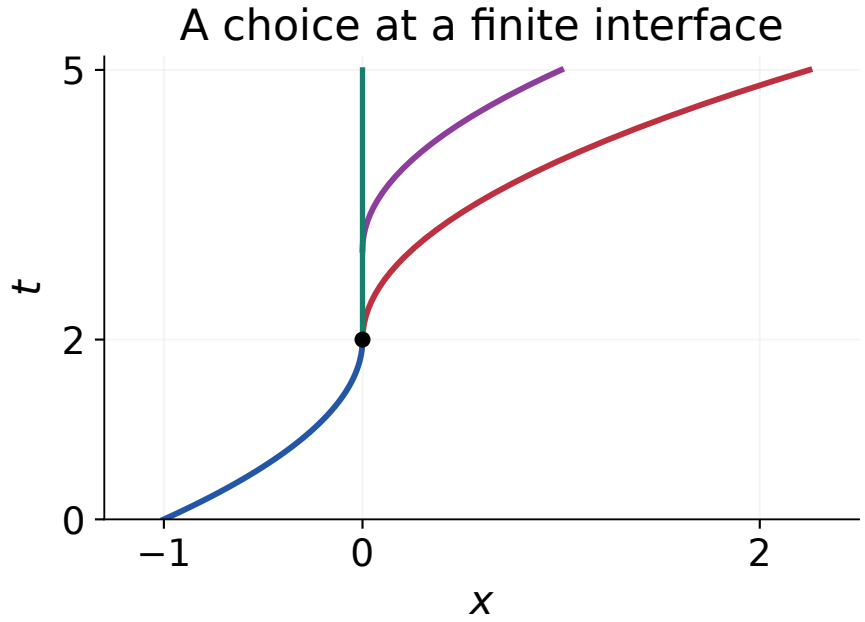
Here, no weak solution exists even near $(0, 0)$.

For the incoming-front data, a weak solution exists exactly up to $t = 2$, when the front arrives.

So, for the continuation theory, we take the entropy inequalities themselves as the solution concept.

Nonuniqueness localised in spacetime





Gargyants–Goritsky–Panov: the disturbances come in from infinity.

Here: the continuations split apart at a finite point and time.

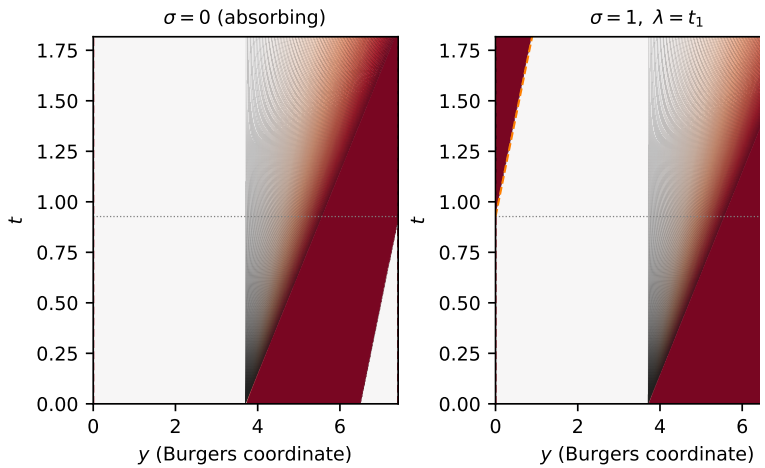
Tychonoff's examples for the heat equation are the classical analogue at infinity.

The Möbius twist

Consider a conservation law on the **Möbius line bundle** over the circle:

$$u_t + \partial_\theta(gu^2) = 0, \qquad g(\theta) = 2\sin(\theta/2).$$

The bundle forces $u(\theta + 2\pi) = -u(\theta)$; flux compatibility forces the same twist on g.



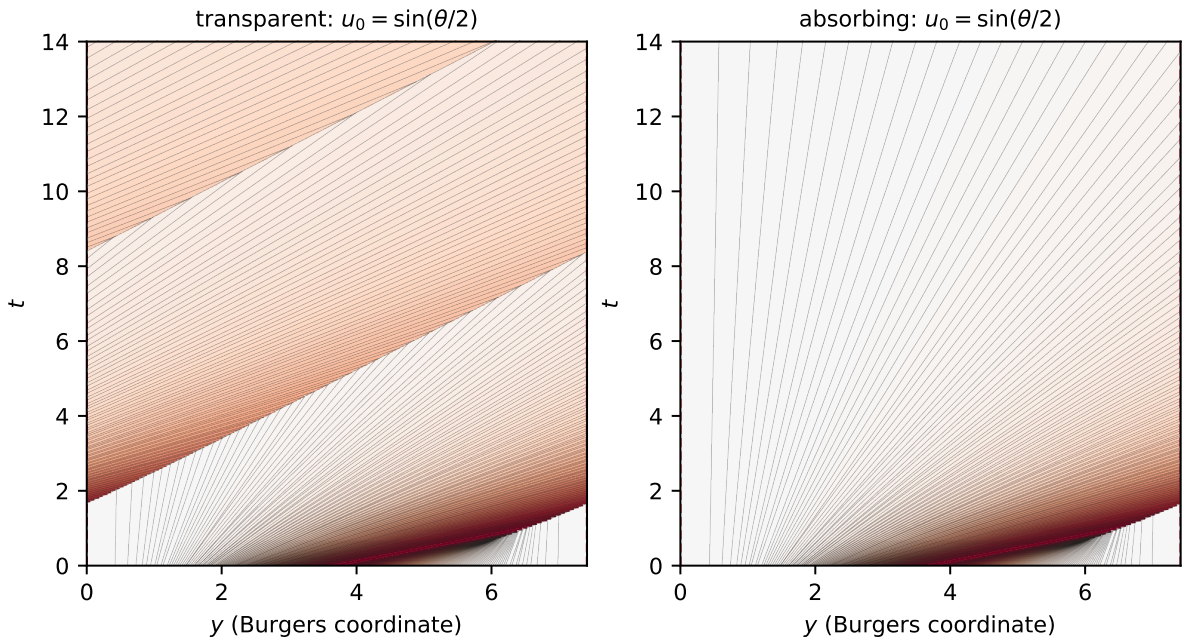
From the same initial box, we can get either absorption or transmission.

Delayed and partial emission are also possible.

The Möbius twist-er

Even for the smooth datum

$$u_0(\theta) = \sin(\theta/2),$$



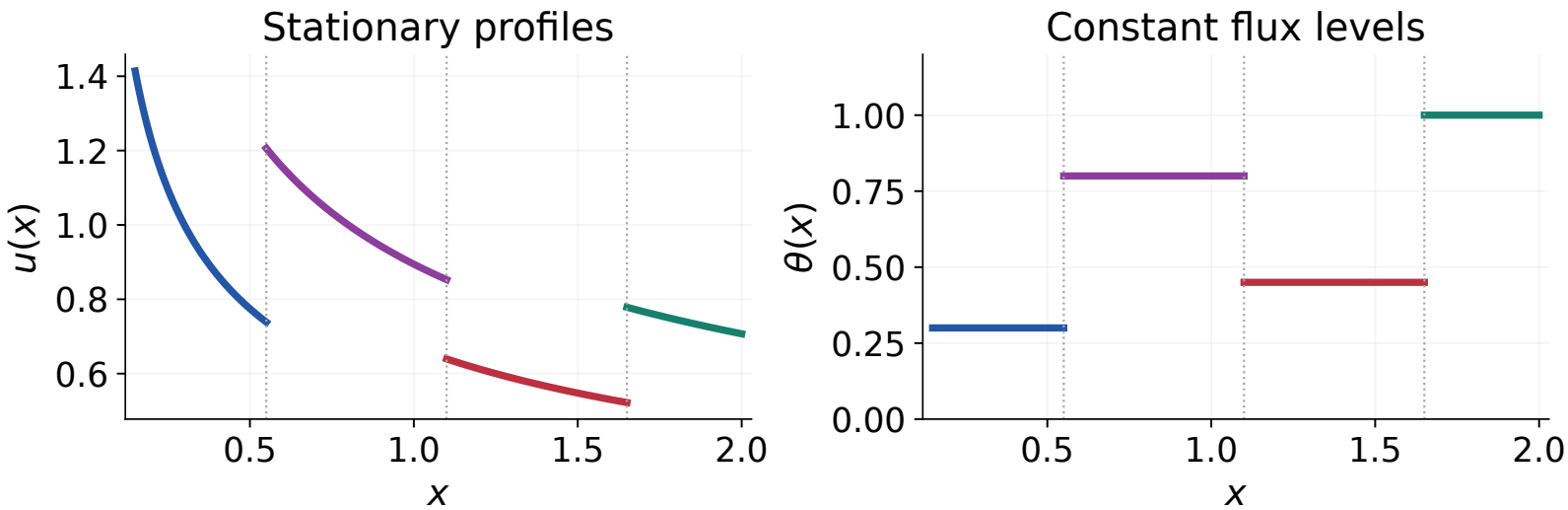
the solution can transmit, absorb, or transmit until a chosen time and then absorb.

So smooth data do not keep us in the locally bounded class forever.

Heterogeneous front tracking

For fluxes of the form $f(x,u) = g(x)h(u)$, we work with the flux level

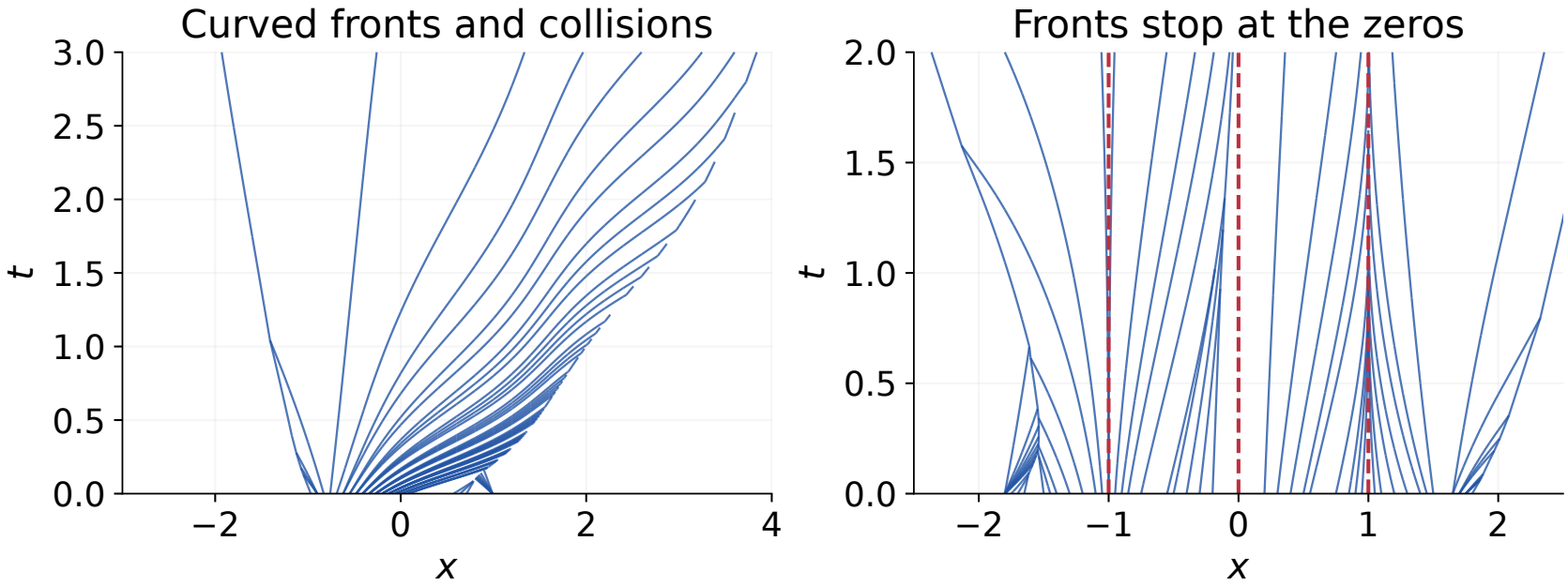
$$\theta = \operatorname{sgn}(u) f(x,u).$$



We approximate θ by finitely many constant levels.

Between fronts, we reconstruct **stationary profiles**, and at each jump, we solve a generalised Riemann problem.

Fronts always collapse



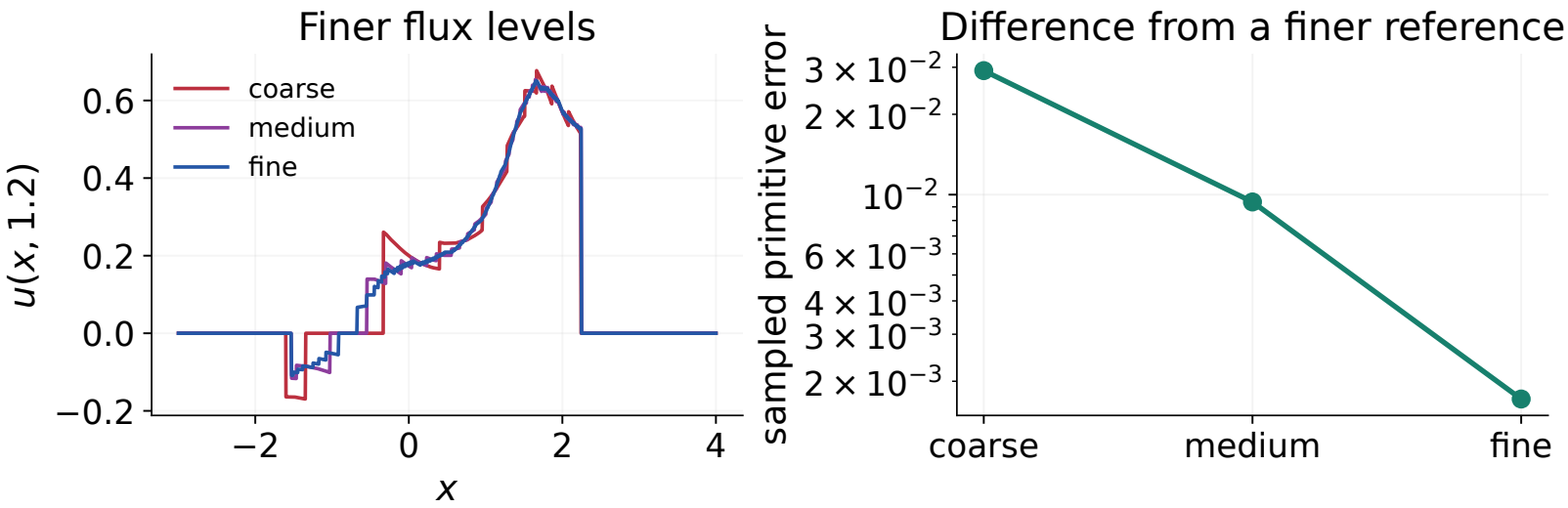
The fronts follow curved Rankine–Hugoniot trajectories.

We resolve collisions as they happen, and apply the interface rule wherever $g = 0$.

Click the figure to play front tracking (15 seconds).

Computed with `hetfront`: $g = 2 + \sin(3x)$ and $g = x(1 - x^2)$, $h = u^2$. Red: zeros of g.

The approximations converge

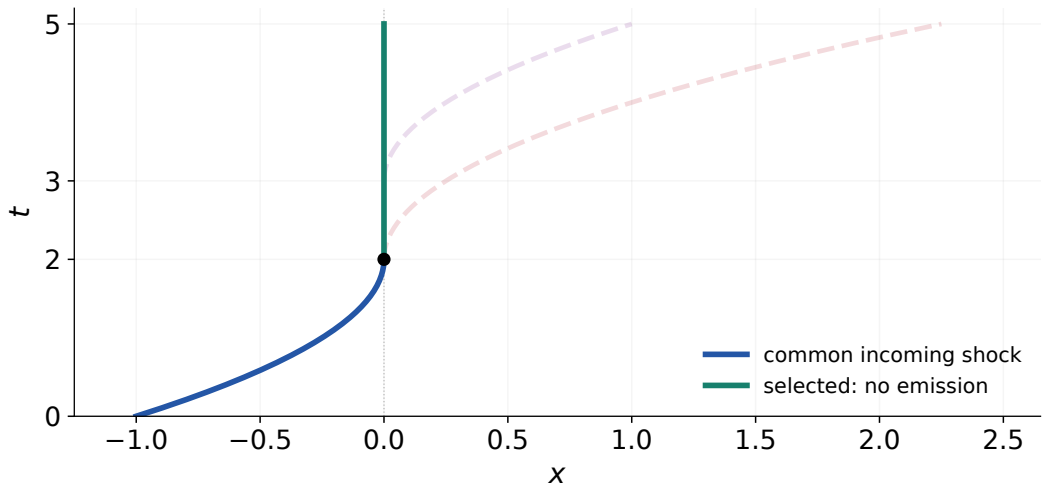


In the paper, we show that front tracking constructs an entropy solution, and that the limit satisfies the interface criterion.

Uniqueness of that limit then comes from the comparison theorem.

Ghoshal–Sylla–Venkatesh, *Failure of uniqueness for scalar conservation laws*, front tracking and stability. Plot: flux-level refinement at fixed spatial discretisation; numerical illustration only.

Entropy and the Lax condition



We impose **no spontaneous inflow** from the singular interface.

For the xu^2 example, this reads

$$u(0+, t) < +\infty,$$

which rules out the positive singular emitted branches.

For Burgers shocks, entropy $\Longleftrightarrow$ Lax. Here, we need entropy **plus** an independent interface rule.

In terms of flux levels: $\theta(z-, t) \geq 0 \geq \theta(z+, t)$.

The Prestige

On a sign interval $I = (a, b)$, the interface rule gives

$$\frac{d}{dt} \int_I |u - v| \, dx \leq Q(a+, t) - Q(b-, t) \leq 0,$$

and hence

$$\int_I |u(x,t) - v(x,t)| \, dx \leq \int_I |u_0(x) - v_0(x)| \, dx.$$

The same initial data now give exactly one admissible continuation.

This holds for locally integrable entropy solutions with bounded flux, under the paper's structural assumptions and the interface criterion.

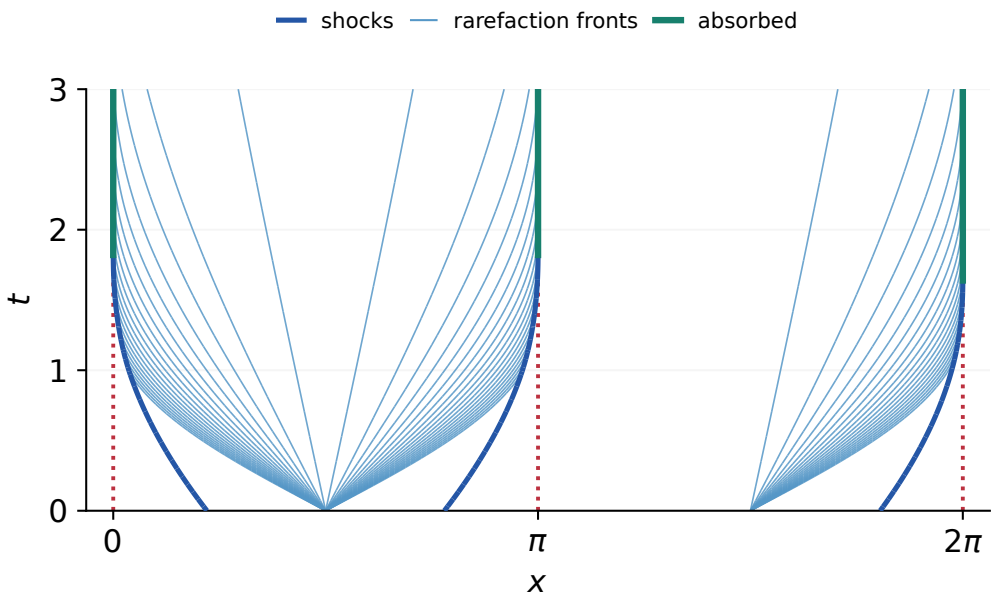
On unbounded intervals, we also have to exclude incoming influence from infinity.

One problem, one solution

Burgers: entropy and Lax agree at shocks.

Here: the Kružkov inequalities alone allow many unbounded continuations.

Once we **add the Lax interface criterion**, uniqueness returns.



$f(x,u) = \sin(x)u^2$

Red dotted lines: $\sin(x) = 0$.

Front tracking constructs the selected limit.

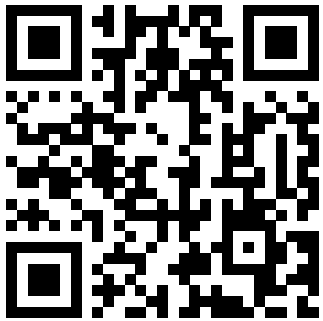
Thank you!

Paper on arXiv



Scan or click

Hetfront code



Scan or click

Möbius playground



Scan or click